Princess Sumaya University for Technology

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Computer Science Department

Princess Sumaya University for Technology
جامعة الأميرة سميّة للتكنولوجيا

Software Engineering 13477

Hotel Reservation System

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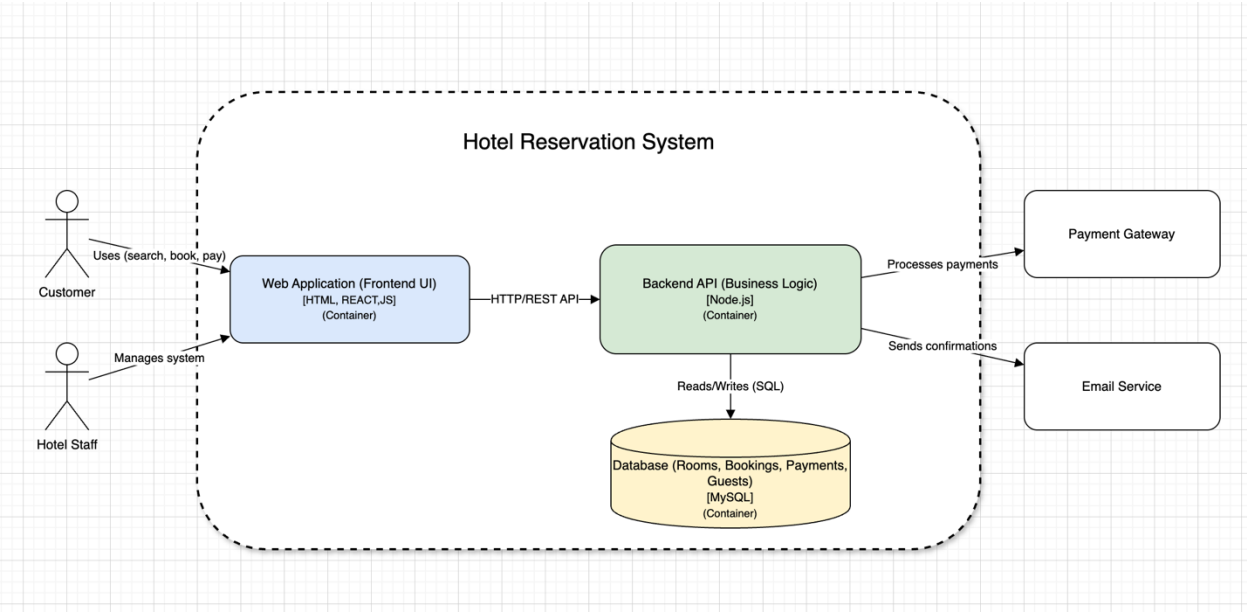
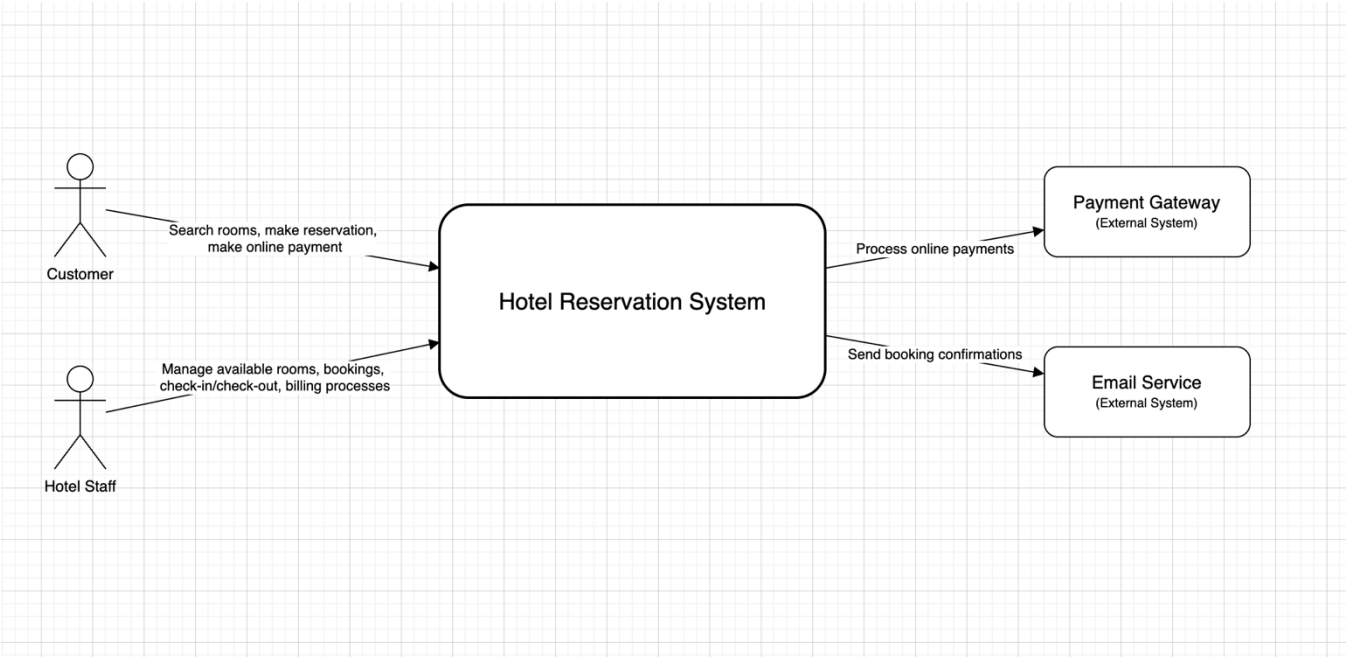
Use Case

A web-based and database-driven system that enables customers to search for available rooms, make reservations, and complete online payments. The system allows hotel staff to manage room availability, guest check-ins and check-outs, and billing processes. It streamlines hotel operations by integrating front-desk management, room scheduling, and payment handling in a single interface. The system maintains records of guests, room types, bookings, and transactions. Integration with payment gateways, email services, and inventory modules ensures real-time booking confirmation and efficient management of resources such as housekeeping and maintenance.

Context Diagram (C4 L1)



Container Diagram (C4 L2)



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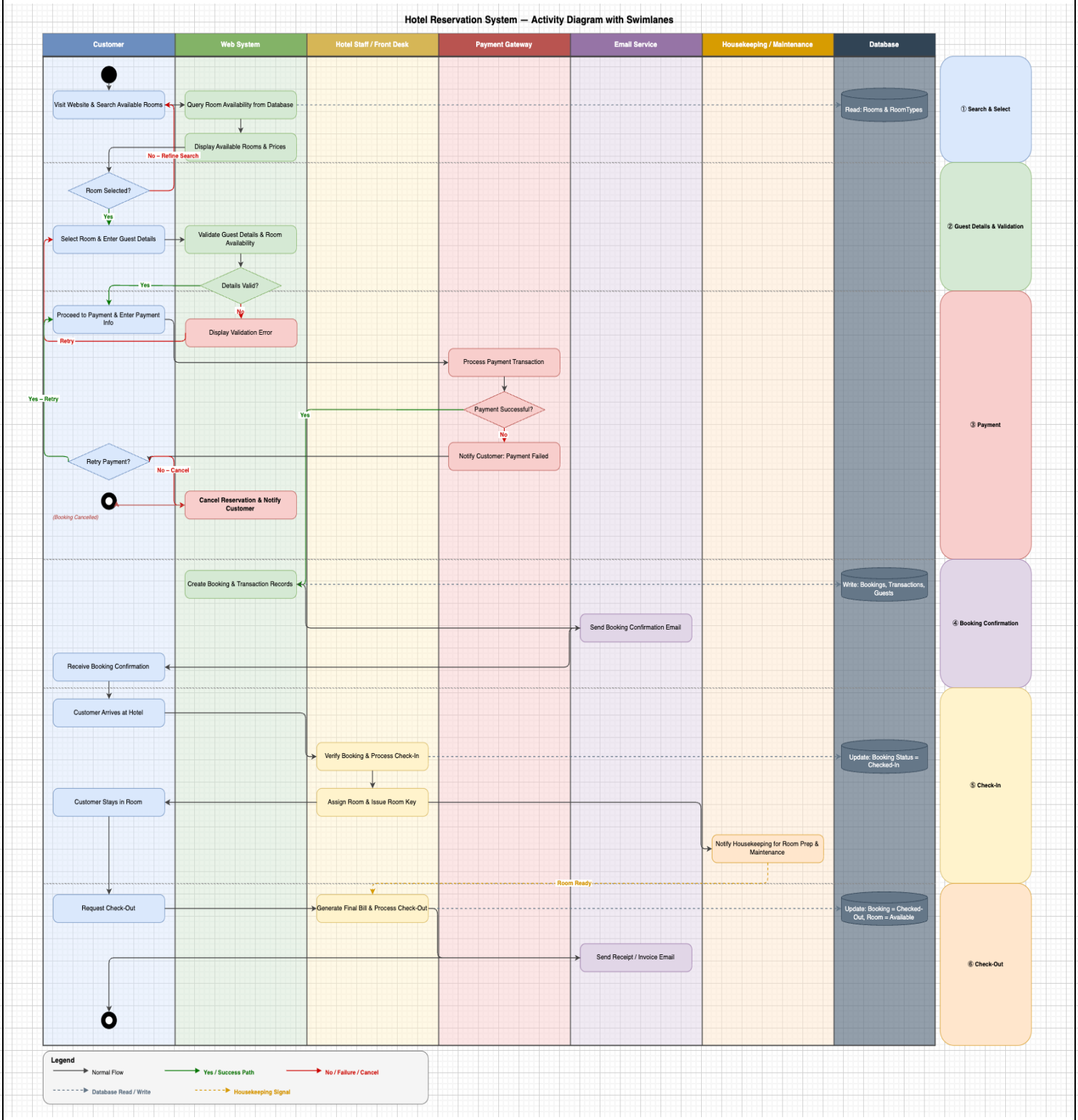
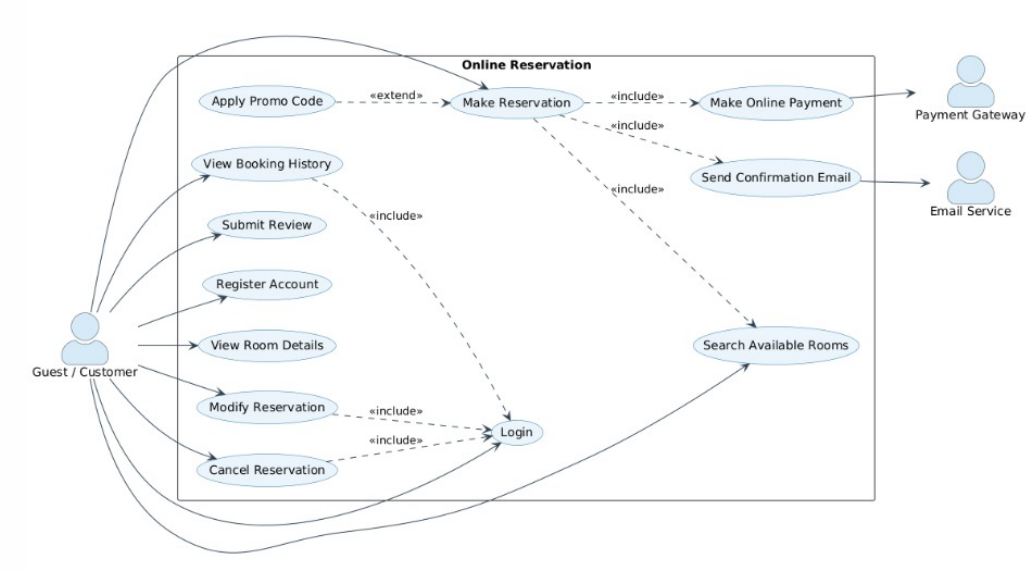
Activity Diagram with Swimlanes

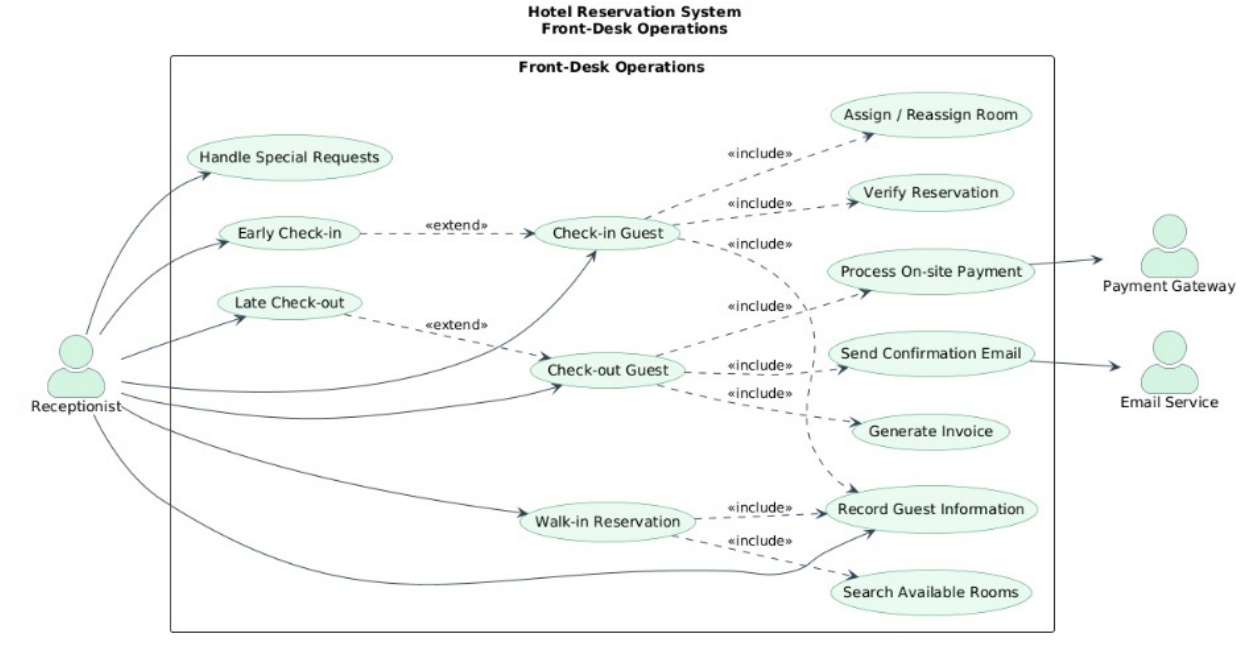


Use Case Diagram

- Online Reservation Use-Case:



- Front Desk Use-Case:



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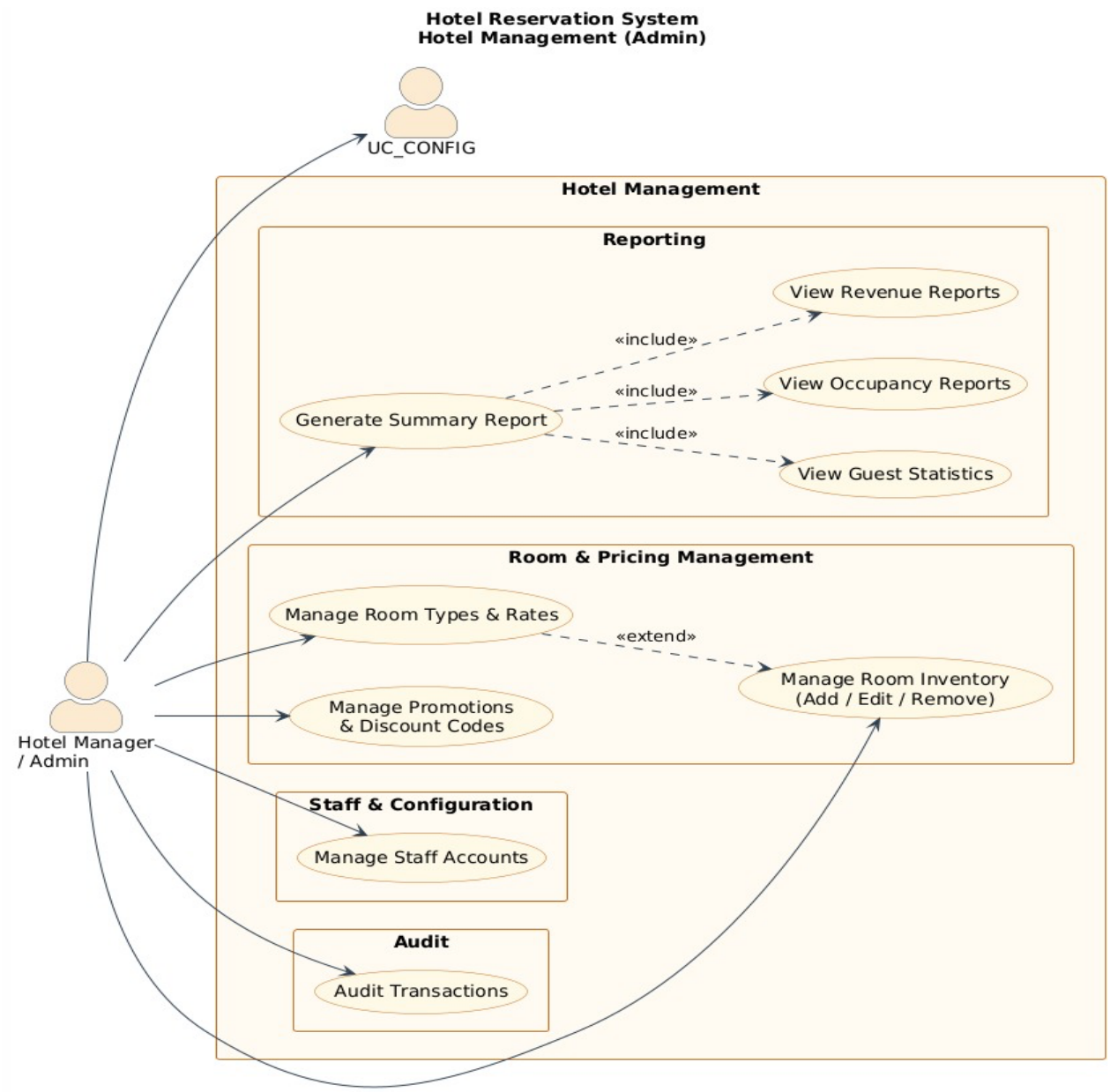
- Hotel Admin Use-Case:



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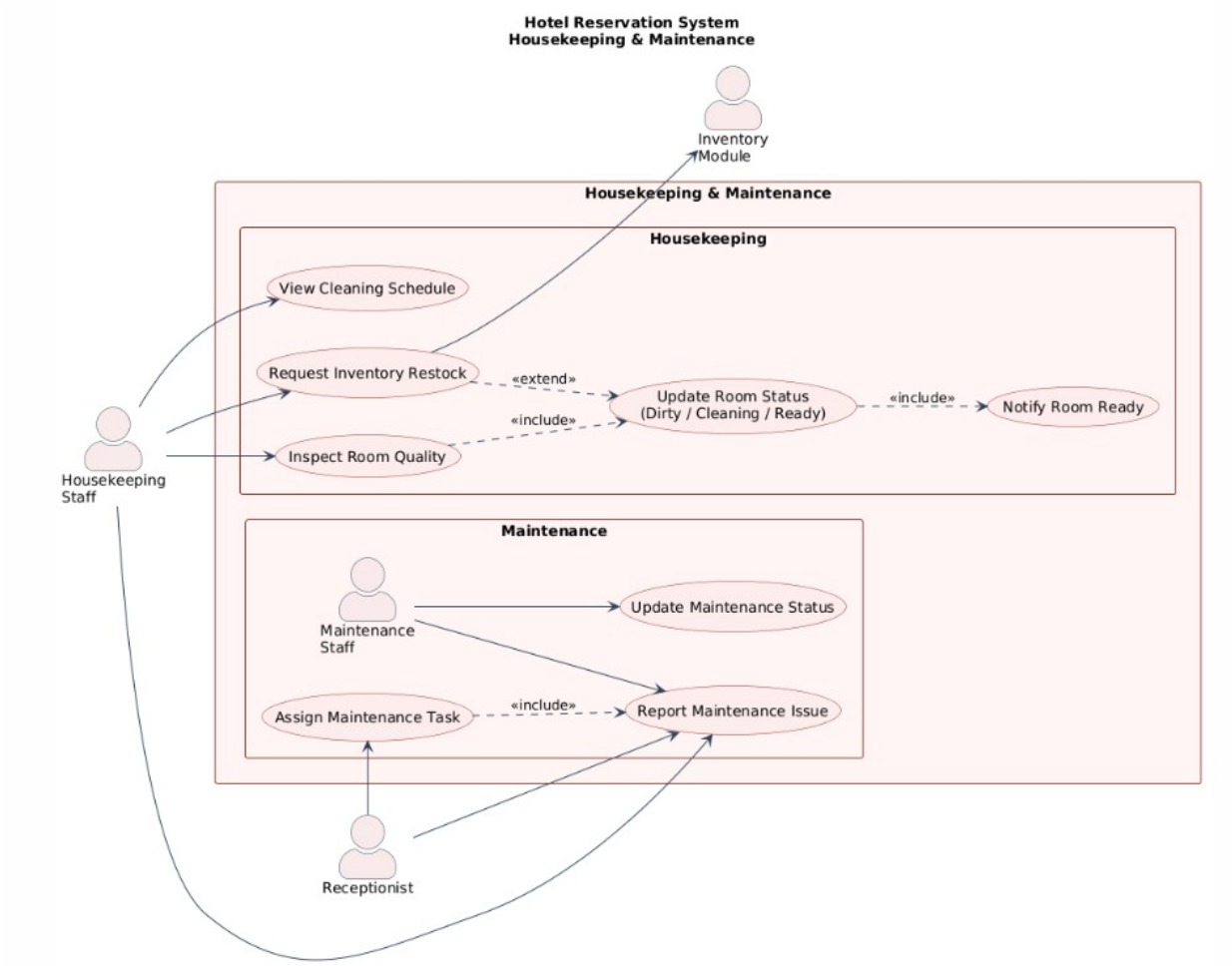
- Maintenance and Housekeeping Use-Case:



Use Case Descriptions (Tabular Format)

Diagram 1 — Online Reservation (Guest-Facing)

#	Use Case	Actors	Description	Relationship
1	**Register Account**	Guest	New user creates a profile with personal details.	—
2	**Login**	Guest	Authenticates user before accessing protected features.	*«include» by Modify, Cancel, History*
3	**Search Available Rooms**	Guest	Filters rooms by date, type, capacity, and price.	*«include» by Make Reservation*
4	**View Room Details**	Guest	Shows photos, amenities, and pricing for a selected room.	—
5	**Make Reservation**	Guest	Books a room for chosen dates and confirms the booking.	*«include» Search, Payment, Email*
6	**Apply Promo Code**	Guest	Optionally applies a discount code to reduce the total.	*«extend» Make Reservation*
7	**Make Online Payment**	Guest, Payment Gateway	Processes booking payment securely via payment gateway.	*«include» by Make Reservation*
8	**Send Confirmation Email**	Email Service	Emails booking confirmation and receipt automatically.	*«include» by Make Reservation*
9	**Modify Reservation**	Guest	Changes dates, room type, or guest count before check-in.	*«include» Login*
10	**Cancel Reservation**	Guest	Cancels booking; triggers refund if policy allows.	*«include» Login*
11	**View Booking History**	Guest	Lists past and upcoming reservations with status.	*«include» Login*
12	**Submit Review**	Guest	Allows post-stay feedback and star rating.	—

Diagram 2 — Front-Desk Operations

#	Use Case	Actors	Description	Relationship
1	**Check-in Guest**	Receptionist	Processes arrival, verifies booking, and assigns room.	*«include» Verify, Assign, Record*
2	**Verify Reservation**	Receptionist	Confirms booking validity, identity, and payment status.	*«include» by Check-in*
3	**Assign / Reassign Room**	Receptionist	Allocates or changes a room based on availability.	*«include» by Check-in*
4	**Check-out Guest**	Receptionist	Finalizes stay, generates invoice, and settles payment.	*«include» Invoice, Payment, Email*
5	**Generate Invoice**	Receptionist	Compiles all charges into a printable bill.	*«include» by Check-out*
6	**Process On-site Payment**	Receptionist, Payment Gateway	Accepts cash or card payment at the front desk.	*«include» by Check-out*
7	**Walk-in Reservation**	Receptionist	Creates an immediate booking for a guest without prior reservation.	*«include» Search, Record*
8	**Search Available Rooms**	Receptionist	Checks real-time room availability for walk-ins.	*«include» by Walk-in*
9	**Record Guest Information**	Receptionist	Captures or updates guest ID, contact, and preferences.	*«include» by Check-in, Walk-in*
10	**Handle Special Requests**	Receptionist	Logs extra needs such as accessibility aids or extra beds.	—
11	**Early Check-in**	Receptionist	Accommodates arrival before standard check-in time.	*«extend» Check-in*
12	**Late Check-out**	Receptionist	Permits stay past standard check-out, possibly with extra charge.	*«extend» Check-out*
13	**Send Confirmation Email**	Email Service	Sends check-out receipt or booking confirmation to guest.	*«include» by Check-out*

Diagram 3 — Hotel Management (Admin)

#	Use Case	Actors	Description	Relationship
1	**Manage Room Inventory**	Hotel Manager	Add, edit, or deactivate rooms including type and features.	—
2	**Manage Room Types & Rates**	Hotel Manager	Define categories and set seasonal or dynamic pricing rules.	*«extend» Room Inventory*
3	**Manage Promotions**	Hotel Manager	Create and control discount codes and flash deals.	—
4	**Manage Staff Accounts**	Hotel Manager	Create, edit, and deactivate staff logins and permissions.	—
5	**Generate Summary Report**	Hotel Manager	Aggregates occupancy, revenue, and guest data into one report.	*«include» Occupancy, Revenue, Guests*
6	**View Occupancy Reports**	Hotel Manager	Tracks room fill rates, peak periods, and availability trends.	*«include» by Summary Report*
7	**View Revenue Reports**	Hotel Manager	Summarizes income by room type, period, and payment method.	*«include» by Summary Report*
8	**View Guest Statistics**	Hotel Manager	Analyzes guest demographics and return rates.	*«include» by Summary Report*
9	**Audit Transactions**	Hotel Manager	Reviews payment logs, refunds, and billing adjustments.	—

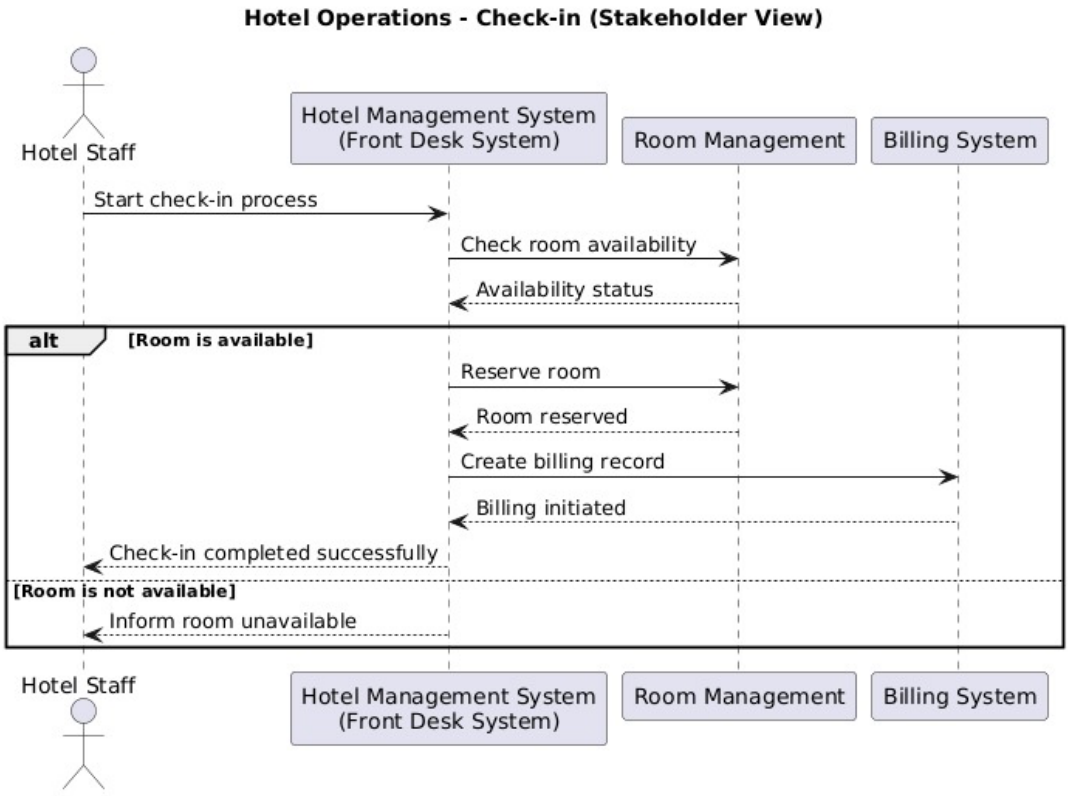
Diagram 4 — Housekeeping & Maintenance

#	Use Case	Actors	Description	Relationship
1	**View Cleaning Schedule**	Housekeeping Staff	Shows daily rooms to clean sorted by priority and check-out.	—
2	**Inspect Room Quality**	Housekeeping Staff	Post-clean quality check before marking room as Ready.	*«include» Update Room Status*
3	**Update Room Status**	Housekeeping Staff	Marks room as Dirty, Cleaning, or Ready in real time.	*«include» Notify Room Ready*
4	**Notify Room Ready**	Housekeeping Staff	Alerts front desk that a room is available for assignment.	*«include» by Update Status*
5	**Request Inventory Restock**	Housekeeping Staff, Inventory Module	Submits request for supplies or amenities to the inventory module.	*«extend» Update Room Status*
6	**Report Maintenance Issue**	Housekeeping Staff, Receptionist, Maintenance staff.	Flags a broken fixture, fault, or safety concern.	*«include» by Assign Task*
7	**Assign Maintenance Task**	Receptionist	Allocates a reported issue to maintenance staff with priority.	*«include» Report Issue*
8	**Update Maintenance Status**	Maintenance Staff	Records progress (In Progress / Resolved) on assigned tasks.	—

Sequence Diagrams

- High-level

Check-In



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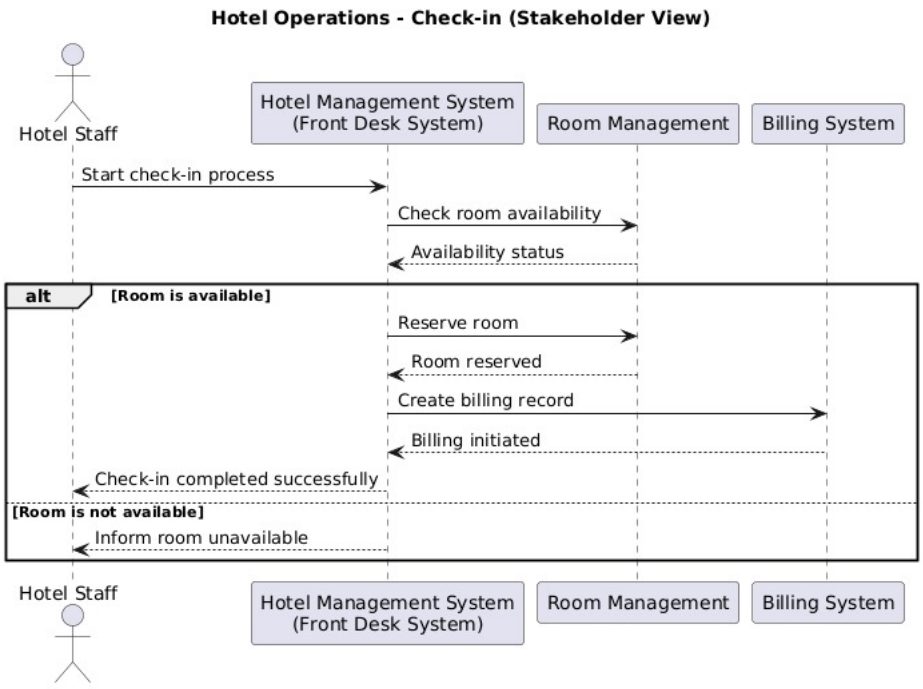
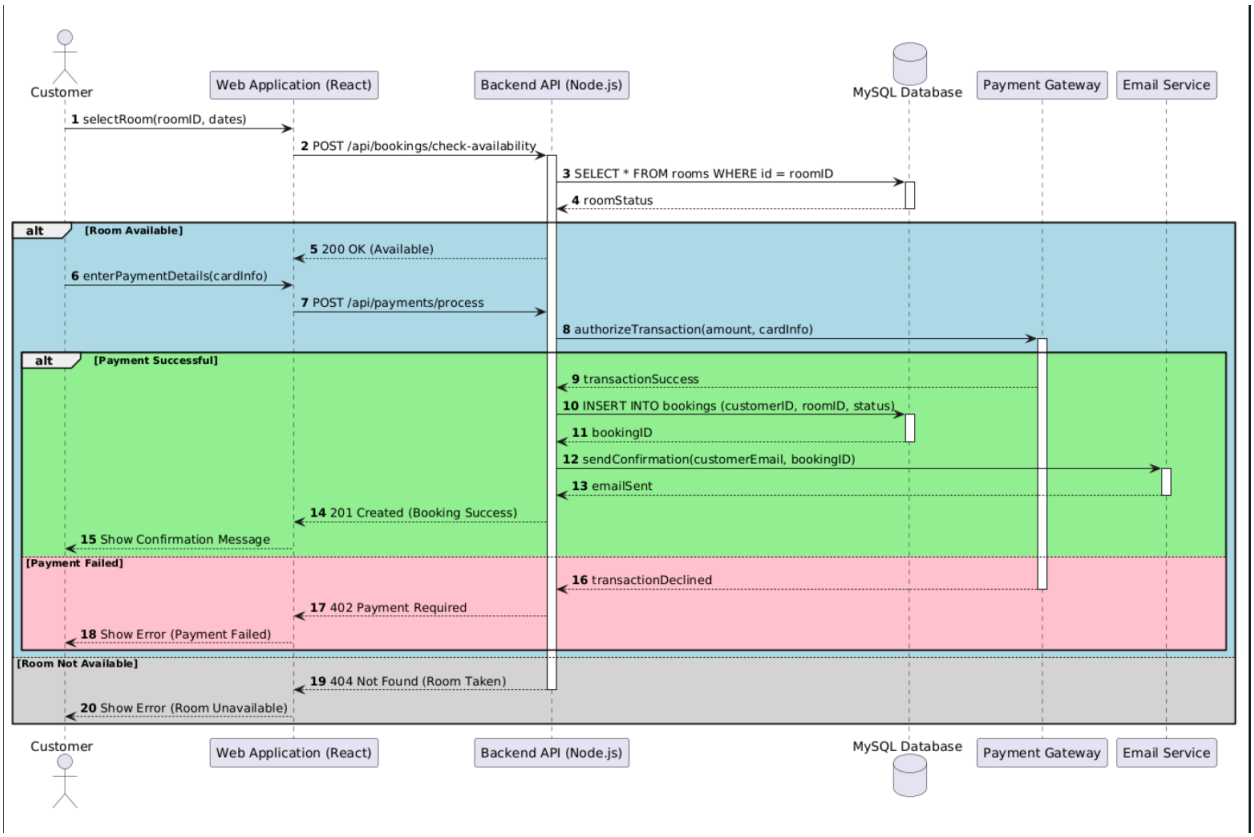
- Check-Out



- Detailed


Customer
Web Application (React)
Backend API (Node.js)
MySQL Database
Payment Gateway
Email Service
1 selectRoom(roomID, dates)
2 POST /api/bookings/check-availability
3 SELECT * FROM rooms WHERE id = roomID
4 roomStatus
alt
[Room Available]
5 200 OK (Available)
6 enterPaymentDetails(cardInfo)
7 POST /api/payments/process
8 authorizeTransaction(amount, cardInfo)
alt
[Payment Successful]
9 transactionSuccess
10 INSERT INTO bookings (customerID, roomID, status)
11 bookingID
12 sendConfirmation(customerEmail, bookingID)
13 emailSent
14 201 Created (Booking Success)
15 Show Confirmation Message
[Payment Failed]
16 transactionDeclined
17 402 Payment Required
18 Show Error (Payment Failed)
[Room Not Available]
19 404 Not Found (Room Taken)
20 Show Error (Room Unavailable)
Customer
Web Application (React)
Backend API (Node.js)
MySQL Database
Payment Gateway
Email Service

Members Roles

- Reef Thneibat → Context Diagram, Container Diagram, Activity Diagram with swimlanes
- Renad Alsayyed → Sequence Diagram (Detailed)
- Omar Tardi → Use-Case diagram, Use-Case descriptions (Tabular Format)
- Mohammad Amaireh → Sequence Diagram (High-Level)

- THE END –